TRIBHUVAN UNIVERSITY

# PATAN MULTIPLE CAMPUS

PATANDHOKA, LALITPUR

**LAB REPORT**

**COMPUTER GRAPHICS (BIT 304)**


**SUBMITTED BY:**

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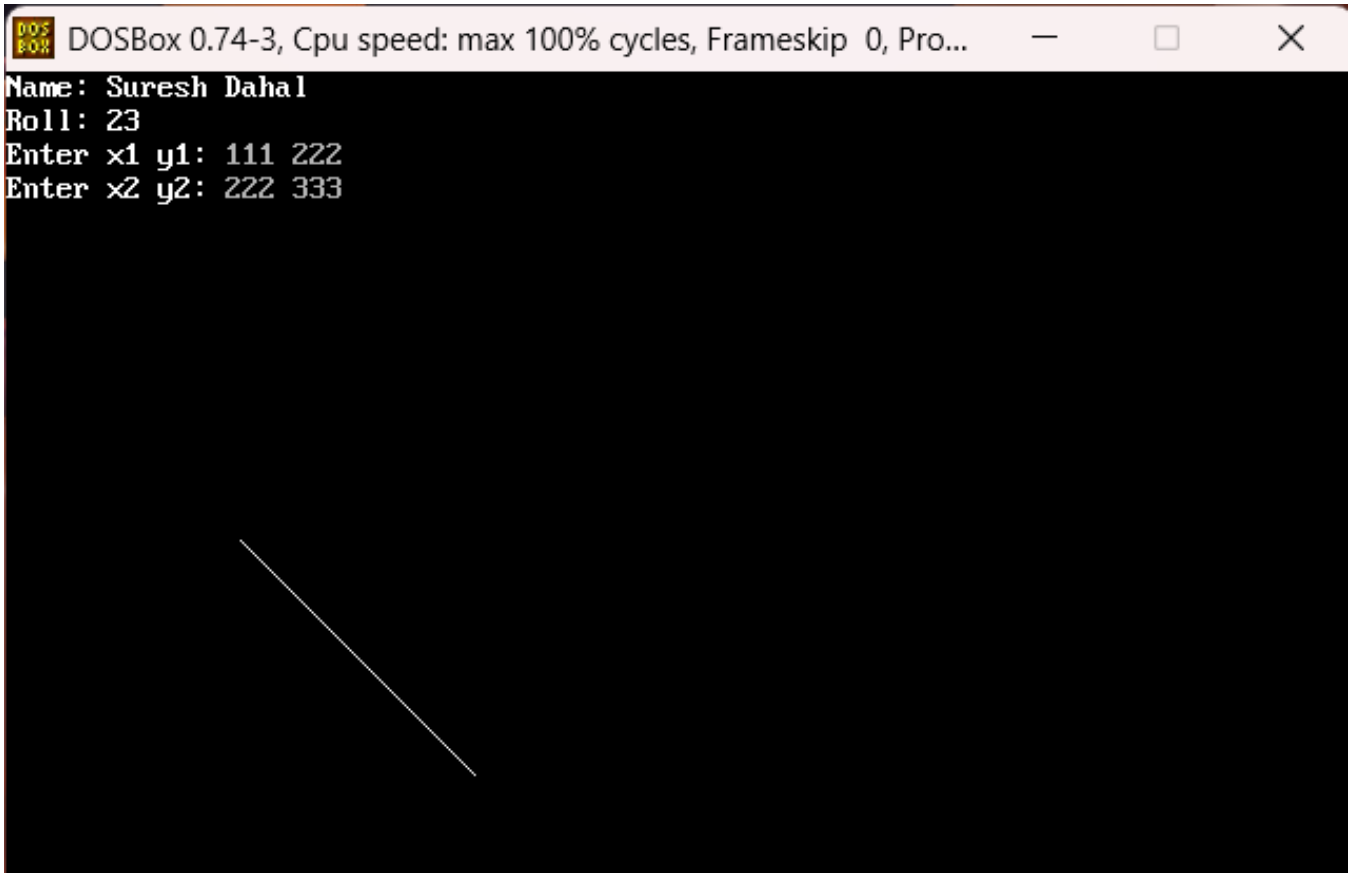
**CLASS: BIT – III/I**
**ROLL: 23**
**DATE: 2082-12-10**

**SUBMITTED TO:**

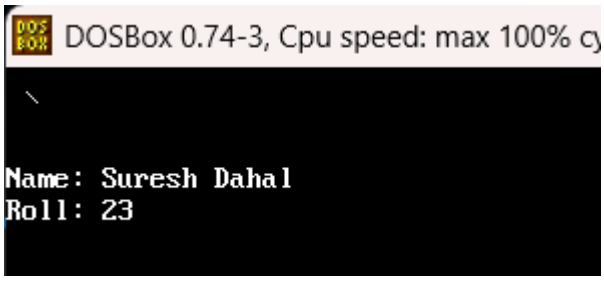
**PHUL BABU JHA**

**CHECKED BY**

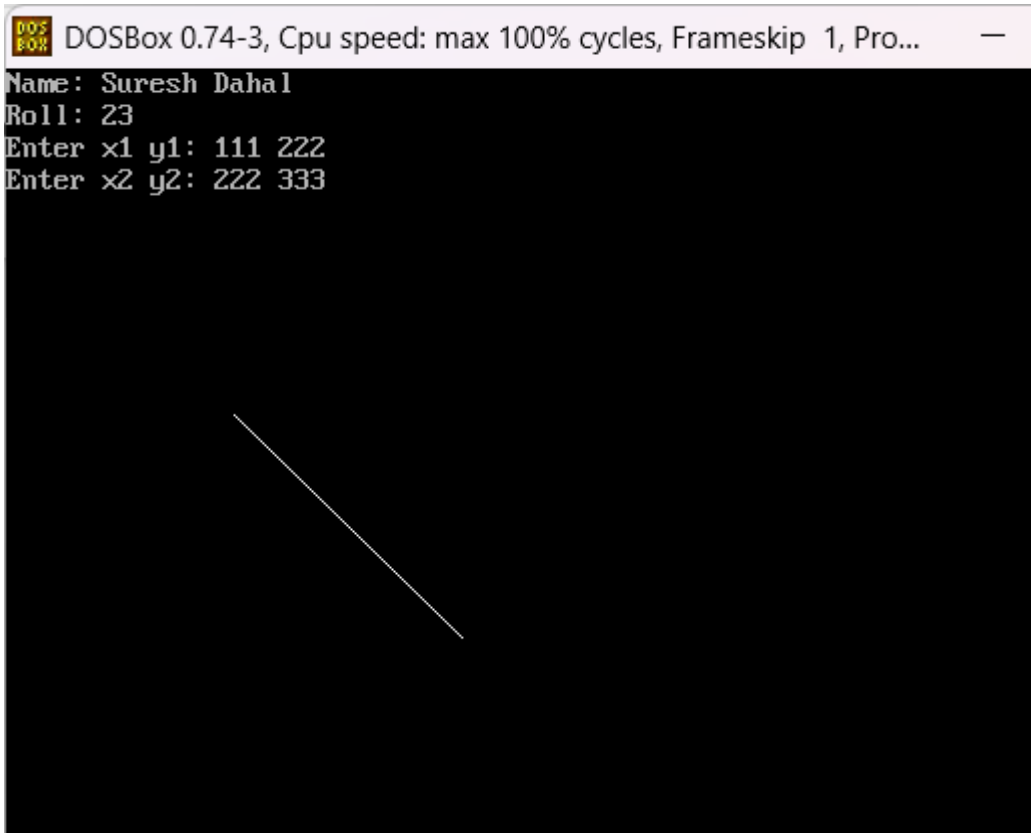
1. **Write a C program for DDA line generation if the user can input the endpoints of the line.**



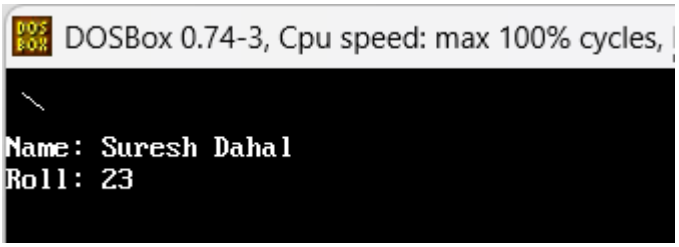
**2. Write a C program for DDA Line generation of Endpoints A(10,12) and B(15,17)**



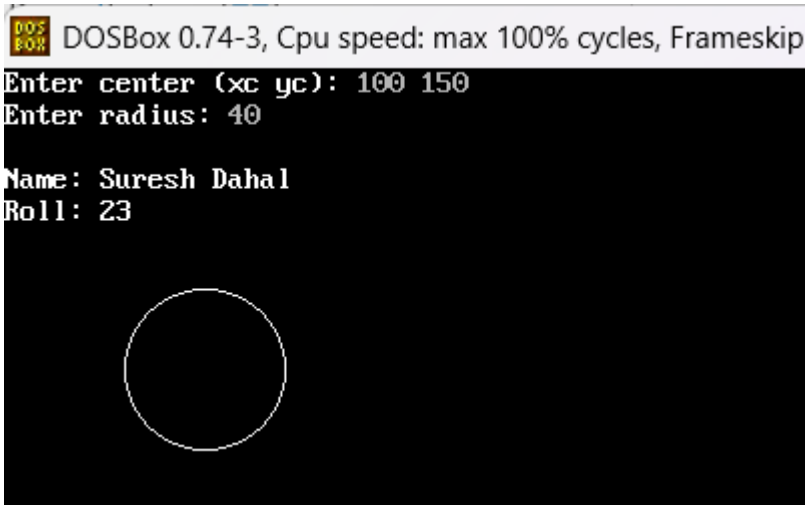
3. **Write a C program for BSA line generation if the user can input the endpoints of the line**



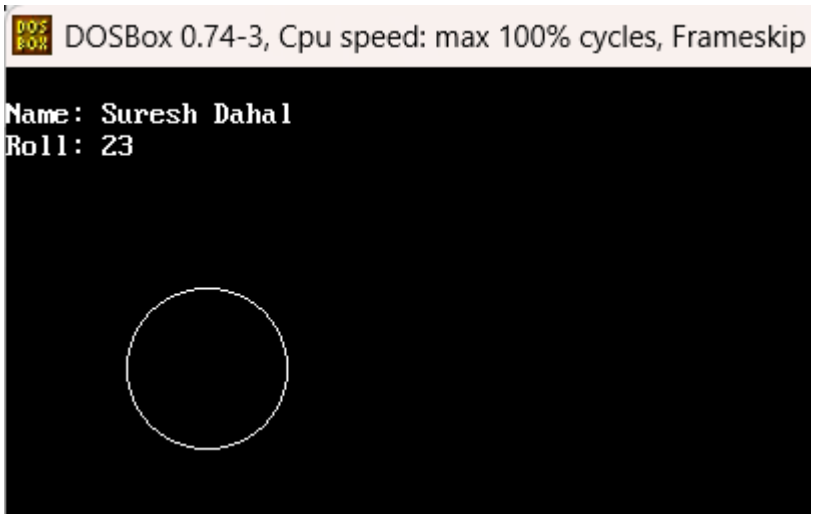
**4. Write a C program for BSA Line generation of Endpoints A(8,12) and B(18,20)**



## 5. Write a C program for generating a BSA circle algorithm

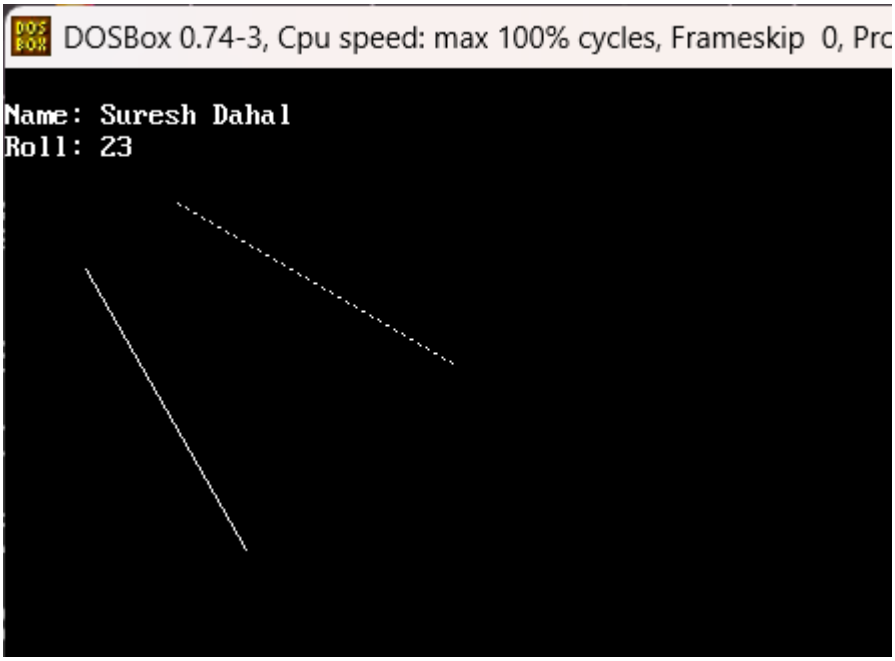


## 6. Write a c program for Generating All points of the Circle by mid-point Algorithm

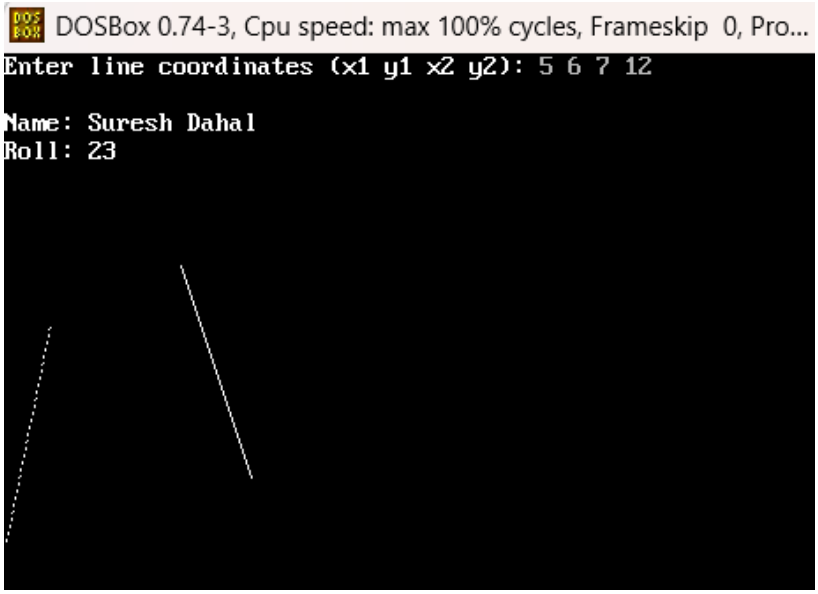


7. **Write a program to rotate the following shape**

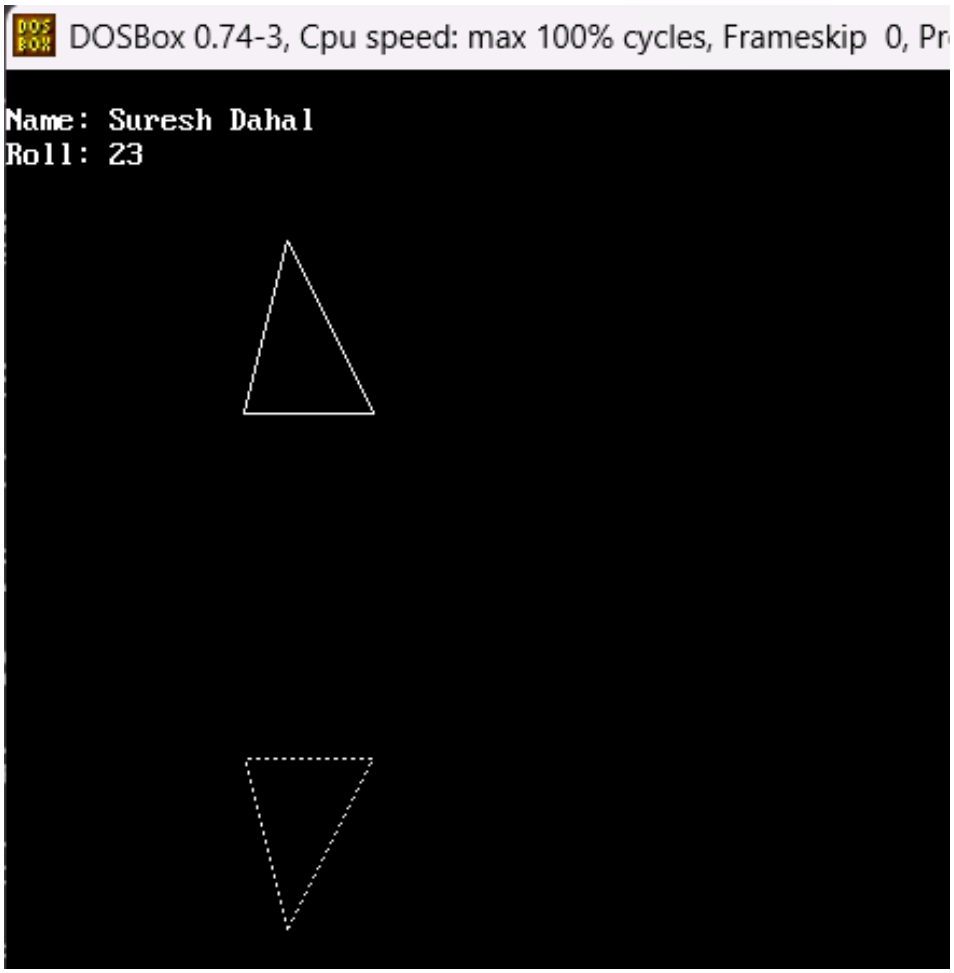
a. **Rotate line AB whose endpoints are A (2, 5) and B (6, 12) about the origin through a 30° clockwise direction.**




b. **Rotate line AB whose endpoints are input by the user about origin through a 30° anticlockwise direction.**

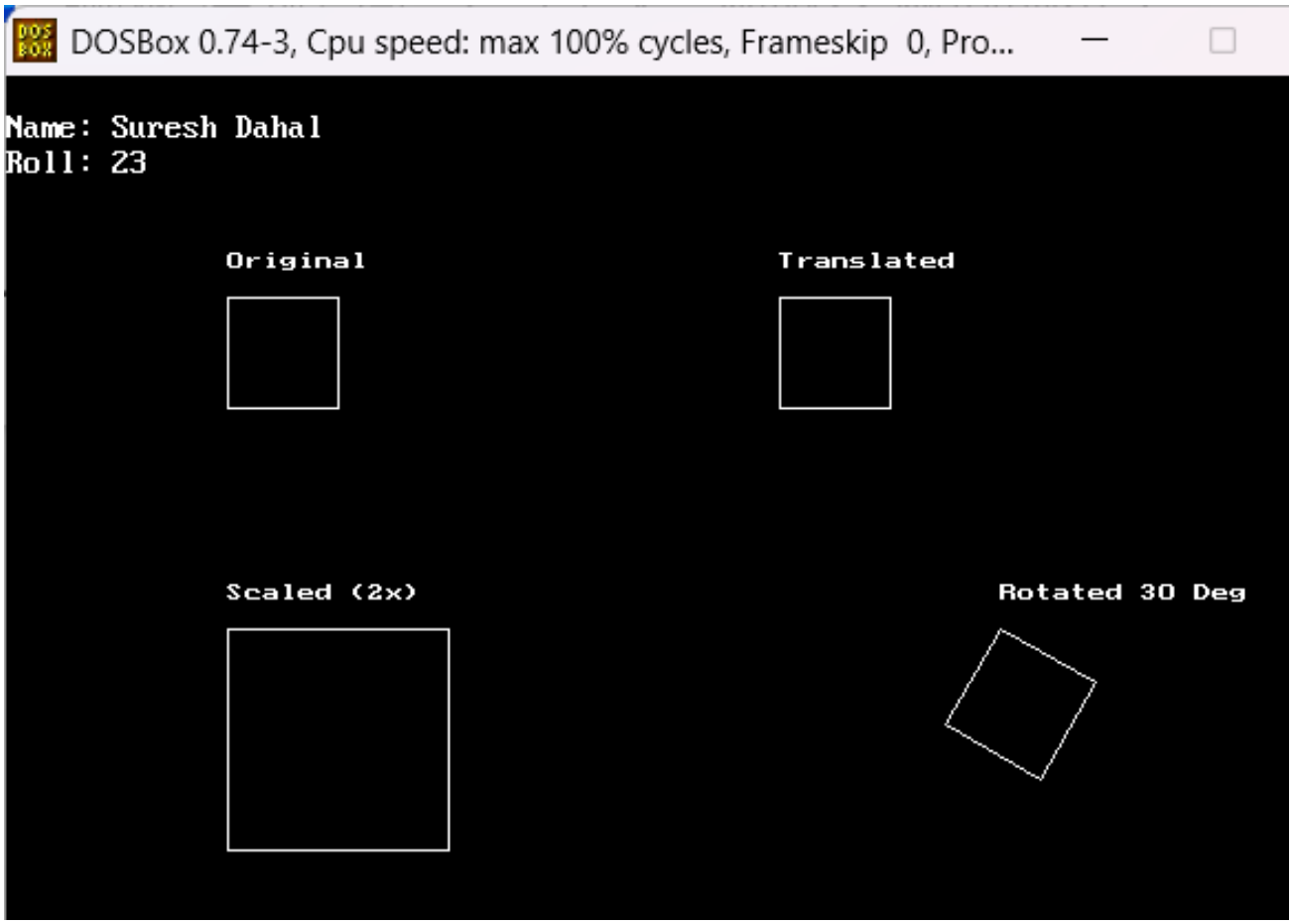


**8. A triangle ABC is given. The coordinates of A, B, and C are given as A (3 4), B (6 4), and C (4 8) Write a program to reflect the position of the triangle i.e., to the x-axis.**

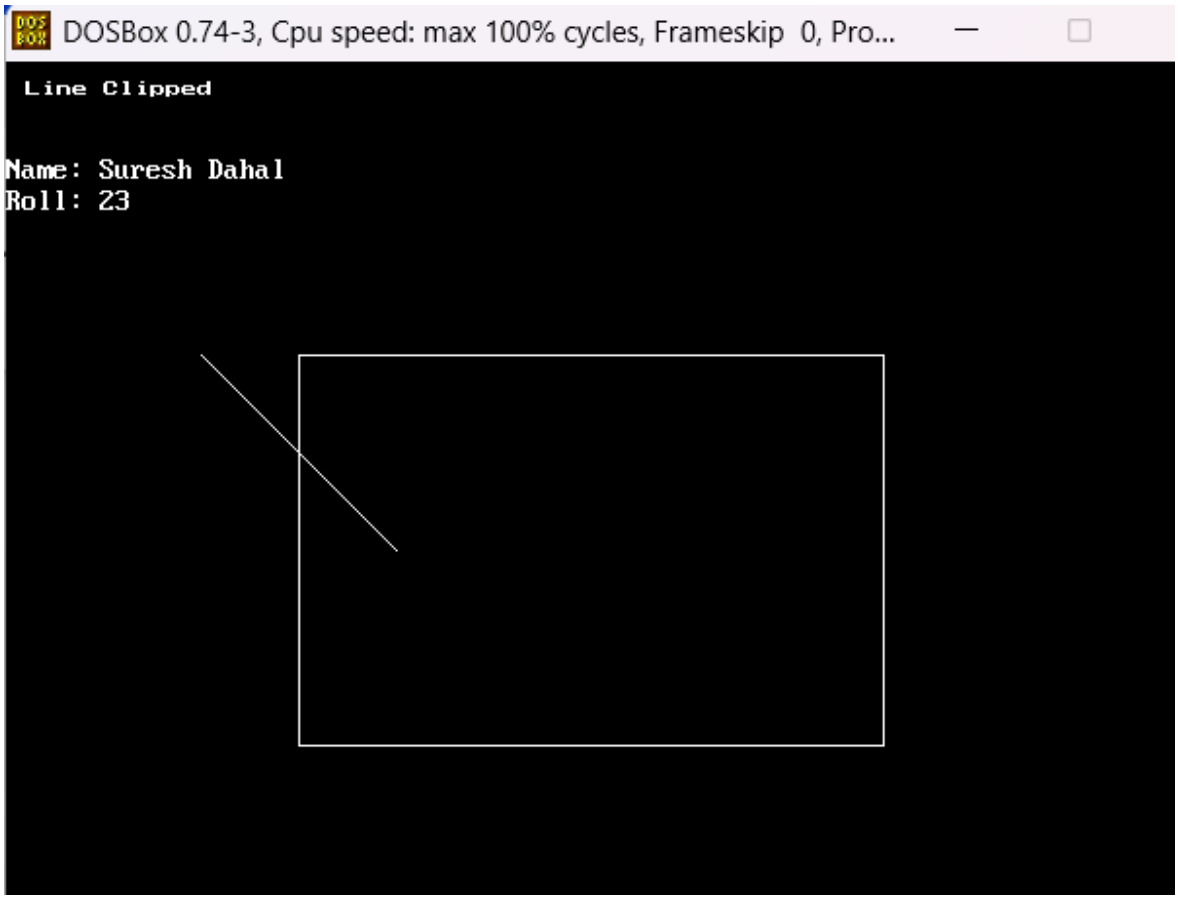


**9. Write a program to perform the following**

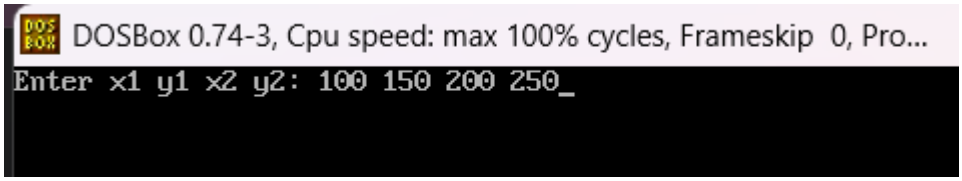
**a. Translation, rotation, scaling**

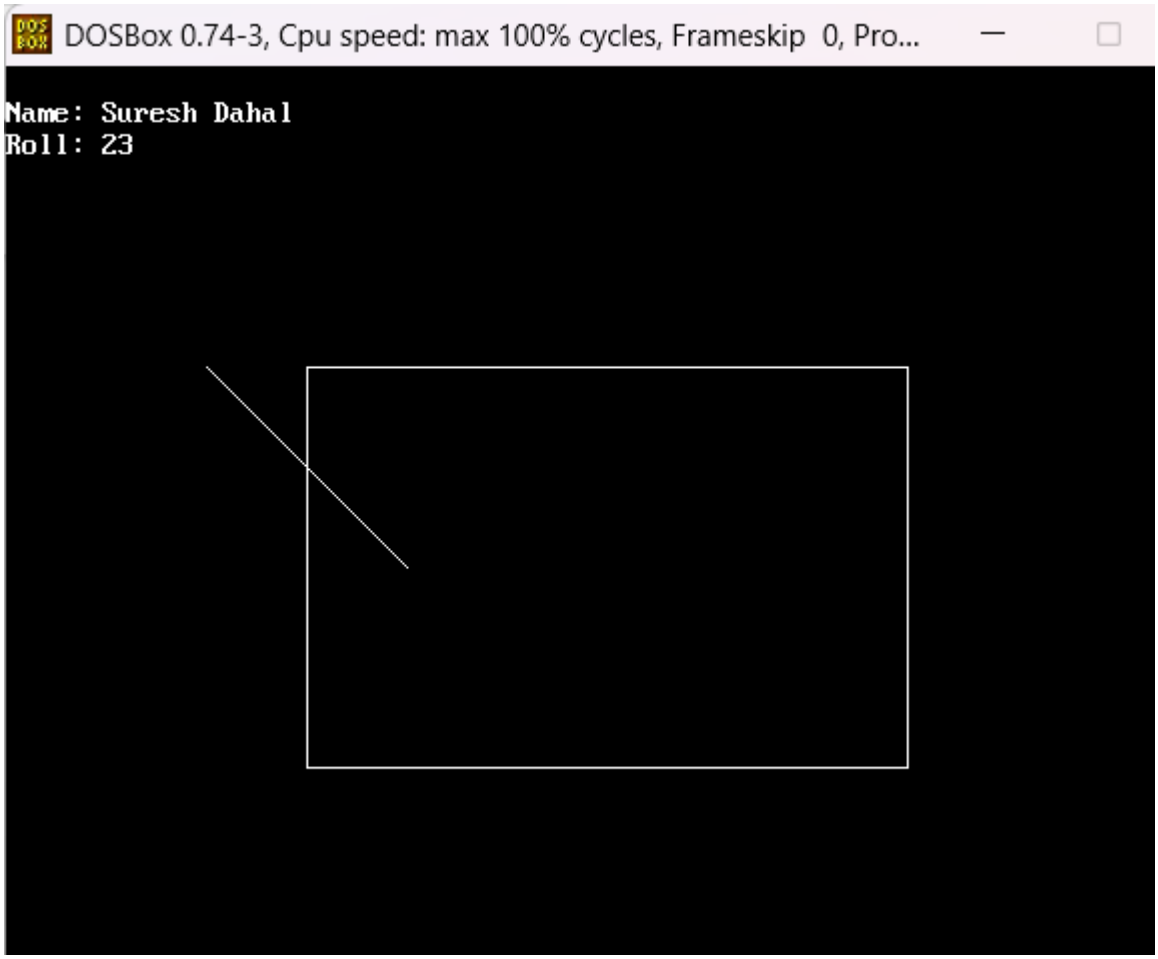


**10. Write a program to perform Line Clipping using Cohen Sutherland Algorithm user can input line coordinates.**

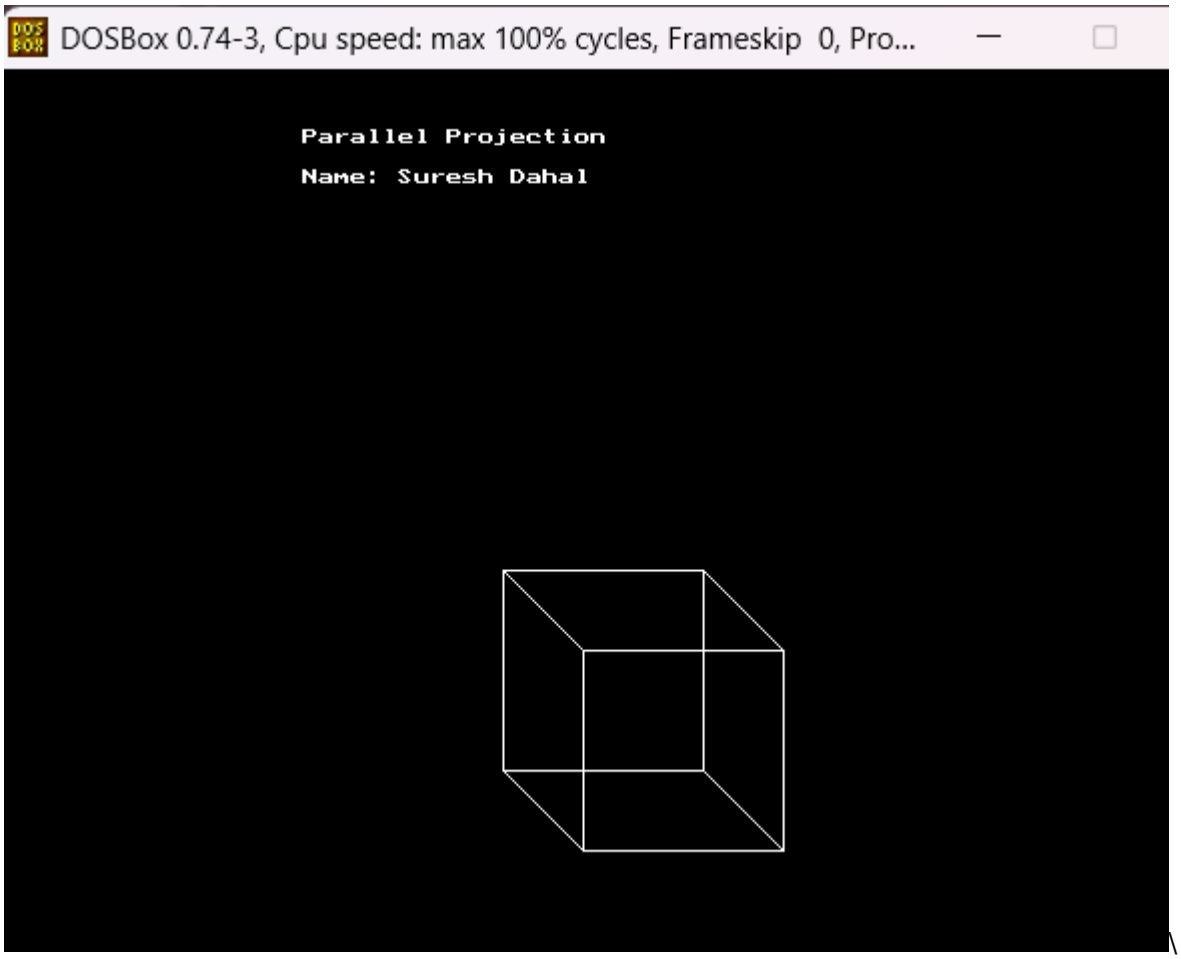


**11.Program to clip a line using the Liang Barsky Method.**

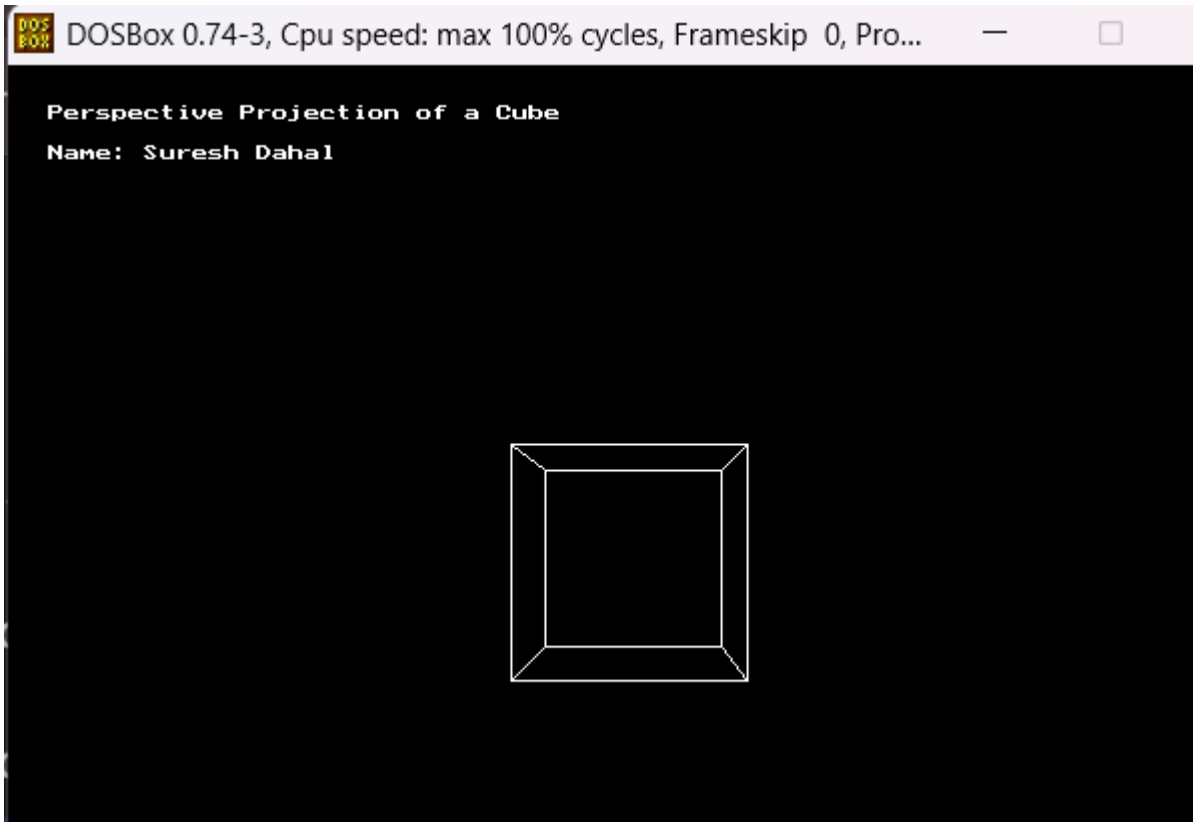
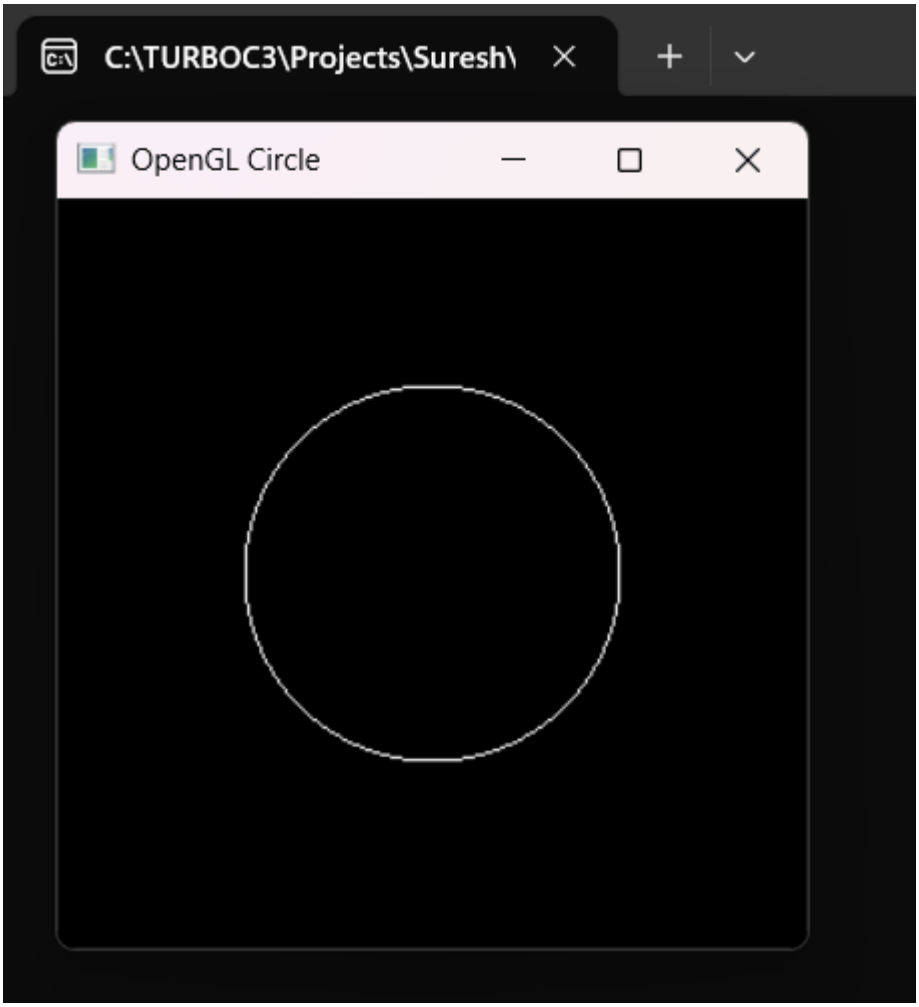


**12.Program to implement Parallel Projection in 3-Dimensions.**



**13.Program to implement Standard Perspective Projection in 3-Dimensions.**



**14.C program to demonstrate drawing a circle using OpenGL**